

Chapter 4

Artifact: monotone sandwich for the network values

The network values decrease to a limit dominating π

Recall the hedging functional and its network approximation,

$$\pi(X) = \inf_{\delta \in \mathcal{H}^u} \rho(X + W(H \circ \delta)), \quad \pi_M(X) = \inf_{\delta \in \mathcal{H}_M} \rho(X + W(H \circ \delta)),$$

where $\mathcal{H}_M \subseteq \mathcal{H}_{M+1}$ are the nested network strategy classes. The following elementary observation reduces the approximation question to a one-sided estimate.

Lemma 4.1. *For every X and every $M \in \mathbb{N}$, one has*

$$\pi_M(X) \geq \pi_{M+1}(X) \geq \pi(X).$$

In particular, $\lim_{M \rightarrow \infty} \pi_M(X)$ exists in $[-\infty, \infty]$ and dominates $\pi(X)$.

Proof. By definition, strategies in $\mathcal{H}_M$ are processes $\delta = (\delta_k)_{k=0, \dots, n-1}$ with

$$\delta_k = F_k(I_0, \dots, I_k), \quad k = 0, \dots, n-1,$$

for network maps F_k , hence each δ_k is $\mathcal{F}_k$ -measurable and δ is adapted. With the nested exhaustion of the network classes one has $\mathcal{H}_M \subseteq \mathcal{H}_{M+1} \subseteq \mathcal{H}^u$, so the infima defining π_M , π_{M+1} and π are taken over increasing sets, which gives both inequalities. The limit exists because the sequence $(\pi_M(X))_{M \in \mathbb{N}}$ is non-increasing and bounded below by $\pi(X)$ in $[-\infty, \infty]$. $\square$